

Probabilistic NBA Game Prediction

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1 Introduction

Our goal for this project is to investigate the probability based relationship between sports and gambling. The legalization of sports gambling has taken society by storm, asking the question: can gambling be broken? Is it possible to beat the sports books and make money in an industry where the house always wins? To easily quantify success, our project will mimic an investment account. Each group member will manage an imaginary budget of ten thousand dollars. With the guidance of each respective mathematical investment strategy, we will observe which model is able to produce the most profit in a one-week stretch. Thus, revealing the 'best' approach to navigating sports betting. These models will be applied to the NBA market, targeting money-line and over-under point total bets. Money-line is a binary win-loss bet, where the better attempts to guess who will win the game. An over-under bet is an a bet targeting the total amount of points scored in a single game. The house will publish a perceived amount of points for both teams to score in a single game. The better then must decided whether more or less points then this benchmark will be scored. The diversity of the bets used correlates with the dynamic market of sports betting. Potentially revealing important information and data based approaches for betting.

2 Background Information

It is important to establish an understanding of gambling odds and book making. There are three odds systems used around the world, but our project utilizes two, the American and Decimal systems. American odds are regularly advertised, commonly shown in TV sports broadcasts and advertisements. While the decimal system is more common overseas, similar to the way the metric system is alien to the United States.

The easiest way to explain American odds is through an example. Suppose you see an advertised boxing match state the odds for fighter one is -100 and +200 for fighter two. But what does this really mean about the two fighters and the potential outcome of the fight? The allocation of a plus and minus signifies a favorite and an underdog. The negative representing the favorite, and the plus or positive for the underdog. In the case of an extremely level match up between fighters, both odds can be displayed as negative. This is a case of 'anyone's game', where either fighter has a good or even an equal chance of winning.

Decimal odds are much more straightforward. Decimal odds represent the proportional winnings a bet may accrue if won. Let us use the example of decimal odds 2.55. Decimal odds can be read on a percent scale, so 2.55 can be translated to 255%. This means, that if this specific bet is won, a better can win a payout that is scaled to 255% the size of the original wager. Additionally, all Decimal odds are positive. Due to the scaling nature of the Decimal odds system, it is logically impossible to achieve a negative payout given a better wins. That said, the more favored a team is to win, the lower the Decimal odds will be. The Decimal system is an easy way to visualize the relationship between risk and expected reward. The larger the Decimal Odds are in comparison to a competitor, the greater a chance that team will lose.

The major appeal of a bet is the expected payout. Given that a sum of money is being wagered, how much can a better profit from correctly predicting the outcome or a feature of a sports game? Decimal odds are the most intuitive to calculate as they serve as scaling factors for the original wager. For example, let us say an NBA basketball game is offering 1.5 decimal odds for your favorite team. You, the better, are interested in putting \$10 down. But how much is \$10 going to win you? The assumed earnings given a correct bet can be found as follows:

Decimal Odds Potential Payout

m: Money paid out after successfully cashing a bet

w: Money wagered

d: Decimal odds provided for said bet

$$m = d * w$$

(Wood, 2025)

As shown above, the decimal odds can simply be multiplied by the original wager to find the potential bet payout. To continue the example above, $\$10 * 1.5 = \15 . Thus yielding a net winnings of \$5 and a total payout of \$15.

The American odds system is less intuitive can is still simple. The potential payout formulas are as follows:

Positive American Odds Potential Payout

m: Money paid out after successfully cashing a bet

w: Money wagered

$$m = \frac{\text{Odds}}{100} * w$$

(Wood, 2025)

Negative American Odds Potential Payout

m: Money paid out after successfully cashing a bet

w: Money wagered

$$m = \frac{100}{|\text{Odds}|}$$

(Wood, 2025)

In addition, odds also reveal information about the probability of winning. The formulas below reveal how a teams winning percentage can be derived from American odds:

Let w = Implied Probability of Winning

Positive American Odds

$$w = \frac{100}{\text{odds} + 100} * 100$$

(Wood, 2025)

Negative American Odds

$$w = \frac{|\text{odds}|}{|\text{odds}| + 100} * 100$$

(Wood, 2025)

Decimal Odds

$$w = \frac{1}{\text{Odds}}$$

(Wood, 2025)

Overround

People love the cliché "the house always wins", but how does this actually materialize? In probability, p is commonly denoted as the probability a chosen event occurs. With $1-p$ being the probability it does not occur. More importantly, $p + (1-p)$ sums to 1. However, book making requires giving the house an edge to guarantee a profit (Wikimedia) . This is called an over round. The over round is the margin above 100% that the two winning probabilities provided for each competing teams sum to (Wikimedia). For example, the NBA advertises a game between the Los Angeles Lakers with odds -149 and the Orlando Magic at +129. The Lakers win probability rounds to about 59% and the Magic to 43.5%. Together they sum to 103.5%. So, the house over round is 3.5%. This is the proportion of the bets placed during the game the house expects to take home after paying out winnings once the game ends.

x: Overround

$$x = 1 - \Sigma(\text{Implied Win Probability})$$

(Wikimedia)

3 Modeling Assumptions

Representing the base level of predictive analytics, this study employs a simulation-based modeling approach to forecast professional basketball game outcomes. Rather than relying solely on closed-form regression models, the methodology uses computational power to simulate between 50,000 and 100,000 virtual games per matchup. This allows probabilistic outcomes to emerge from historical performance data while preserving the inherent randomness of sports competitions.

4 Probabilistic Game Outcome Simulation Model

The proposed model is a hybrid probabilistic framework combining Monte Carlo simulation techniques with an Elo rating system. The first component simulates game scores using historical offensive and defensive statistics, while the second estimates win probabilities using dynamically updated team strength ratings. Together, these methods provide a comprehensive prediction of matchup outcomes by capturing both performance variability and overall team quality.

4.1 Assumptions

To maintain consistency and reduce uncontrolled variability, the following assumptions were adopted:

- Teams maintain relatively stable performance trends following the trade deadline.
- Major injuries are not explicitly modeled; rotations are assumed to reflect typical starter usage.
- Strategic resting or benching for easy matchups is ignored.
- Historical team performance is treated as a proxy for current team strength.

These assumptions allow the simulations to isolate statistical tendencies rather than unpredictable real-world disruptions.

4.2 Statistical Performance Simulation Method – Normal Distribution Team Scoring Model

The first approach, referred to as the **Normal Distribution Team Scoring Model**, simulates game outcomes by modeling team scoring as a random variable drawn from historical performance distributions. This method assumes that points scored by a team follow an approximately normal distribution characterized by a mean (average scoring) and standard deviation (variability in scoring).

For each team:

- **Average points scored** represents offensive strength.
- **Average points allowed** represents defensive strength.
- **Standard deviation** captures game-to-game scoring variability.

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Simulated game scores are generated using normal probability distributions centered on these parameters. Defensive effects are incorporated by subtracting a randomly generated defensive allowance value from the opponent's offensive score. The team with the higher simulated score is declared the winner.

Why This Approach Is Different

This method focuses on how a team's performance can vary from one game to the next, rather than just looking at overall season performance. In real life, a team does not score the same number of points every game — some nights they may shoot very accurately and score more than usual, while other nights their shooting may be poor, resulting in fewer points. Similarly, a team's defense can sometimes fail, allowing the opponent to score more than expected. The speed of the game, or “pace,” also affects scoring: faster games tend to have higher point totals, while slower games usually have fewer points. By simulating these kinds of variations, the model can show how individual games might turn out differently even when two teams are considered equally strong overall. Essentially, it captures the natural unpredictability of sports. Why a weaker team can sometimes win, and why a stronger team can have an off night.

4.3 Elo Rating Simulation and other methods — Dynamic Team Strength Rating Model

The second approach, called the **Dynamic Team Strength Rating Model**, uses an Elo rating system to quantify overall team quality. Each team is assigned a numerical rating that updates after every game based on expected versus actual results.

Win probability between two teams is calculated using the logistic Elo formula:

$$P(\text{Team A wins}) = \frac{1}{1 + 10^{(R_B - R_A)/400}}$$

where R_A and R_B are the Elo ratings of the two teams.

Game outcomes are simulated repeatedly using these probabilities, producing estimated win frequencies.

Additional Predictive Models

To improve predictive accuracy, several additional statistical and machine learning models are incorporated alongside the Elo rating system.

Monte Carlo Score Simulation Model This model simulates individual game scores using probabilistic sampling based on historical offensive production. Each team's expected score is modeled using a normal distribution centered on its average points per game. The difference in Elo ratings between teams is used as a small adjustment to shift the expected scoring output.

Thousands of simulated games are generated, and the proportion of simulations in which one team scores more points than the opponent represents the estimated probability of winning.

Logistic Regression Prediction Model A logistic regression model is trained using historical game outcomes and team statistics. The model estimates the probability that the home team wins based on differences between teams in several key metrics:

- Win percentage
- Average points scored
- Average points allowed
- Season points per game (PPG)

These statistical features are used as predictors in a logistic probability function, allowing the model to estimate how likely a team is to win based on historical trends.

Neural Network Prediction Model To capture more complex relationships between team statistics and game outcomes, a feed-forward neural network model is also implemented. This model uses multiple hidden layers to learn nonlinear patterns that may exist in historical data.

The same statistical inputs used in the logistic regression model are provided to the neural network, but the model can learn more complex interactions between these variables. This allows the model to identify patterns that simpler linear models may miss.

Why This Approach Is Different

Unlike the statistical scoring model, the Elo method captures **long-term consistency and relative team strength** rather than individual game scoring variability. It reflects how strong a team is overall, independent of specific scoring distributions.

Additionally, the inclusion of machine learning models allows the framework to capture statistical relationships between team performance metrics and game outcomes, while the Monte Carlo simulations model randomness and scoring variability in individual games.

4.4 Key Differences Between Predictive Approaches

Together, these methods complement each other. The statistical scoring model captures game-level randomness, the Elo model measures overall team strength, and the machine learning models identify statistical patterns in historical performance. Combining these approaches allows the simulation framework to account for both randomness in individual games and systematic differences in team quality.

The updated simulation framework integrates four complementary predictive methods to estimate NBA game outcomes:

1. **Monte Carlo Score Simulation Model**
2. **Elo + Offensive Strength Adjustment Model**
3. **Logistic Regression Model**
4. **Neural Network Model**

This hybrid approach captures both short-term game variability and long-term team performance trends. By combining these methods, the model produces more robust predictions for individual matchups.

4.5 Assumptions

To maintain consistency and isolate statistical trends, the following assumptions are adopted:

- Teams maintain relatively stable performance trends following the trade deadline.
- Major injuries are not explicitly modeled; rotations reflect typical starter usage.

- Strategic resting or benching for easier matchups is ignored.
- Historical team performance serves as a proxy for current team strength.
- Monte Carlo simulations assume scoring follows a normal distribution with variance approximating real-game variability.
- Logistic regression and neural network models assume historical patterns of team statistics can predict future outcomes.

These assumptions allow the model to focus on probabilistic outcomes rather than unpredictable real-world disruptions.

4.6 Dataset Preparation

The simulation relies on three primary datasets:

1. **Games.csv** – Contains historical NBA game results with home/away scores.
2. **Team Totals.csv** – Contains season-level statistics, including total points, games played, and derived points per game (PPG).
3. **TeamStatistics.csv** – Contains game-level team statistics for neural network training, including scores, wins, and other box-score metrics.

Data preprocessing steps include:

- Converting numeric columns (e.g., points, games played, scores) to numeric values.
- Removing incomplete or missing records.
- Mapping historical team names to modern franchises for consistency.
- Computing derived statistics, such as points per game (PPG), average points scored, average points allowed, and win percentages.

4.7 Monte Carlo Score Simulation Model

This method models game outcomes by simulating points scored as random variables drawn from normal distributions based on historical PPG:

- Offensive PPG for each team is adjusted slightly according to Elo rating differences.
- Random scores are generated from distributions representing scoring variability.
- Repeated simulations produce an estimated probability of one team winning.

Purpose: Captures game-to-game variability and allows weaker teams to occasionally outperform stronger teams, reflecting the unpredictable nature of sports.

4.8 Elo + Offensive Strength Model

This method builds upon traditional Elo ratings by adjusting for offensive efficiency:

- Elo ratings are updated after each historical game based on expected vs. actual outcomes.
- Each team's PPG relative to league average scales the base Elo win probability.

Purpose: Combines long-term team strength with offensive efficiency to estimate win probabilities, capturing overall team quality.

4.9 Logistic Regression Model

A supervised learning approach using historical team statistics:

- Features include differences in win percentage, points scored, points allowed, and season PPG between teams.
- The model estimates the probability that the home team wins based on historical trends.

Purpose: Leverages patterns in historical team performance to predict outcomes beyond simple scoring distributions.

4.10 Neural Network Model

A feed-forward neural network trained on game-level statistics for nonlinear pattern recognition:

- Inputs mirror those used in logistic regression: differences in win percentage, points scored, points allowed, and PPG.
- Architecture includes multiple hidden layers to capture complex relationships between features and outcomes.
- Standardized inputs improve model performance and training stability.

Purpose: Captures nonlinear and complex interactions between team statistics and game outcomes, providing an additional layer of predictive power.

4.11 Combined Simulation Procedure

For each matchup:

1. Win probabilities are computed using all four models.
2. The predictions are averaged to produce an **overall probability** of one team winning.
3. Results are stored for further analysis or visualization.

This ensures that both stochastic variability and consistent team performance trends are considered.

4.12 Visualization

Predicted outcomes for each matchup are visualized using bar charts:

- Horizontal bars display predicted wins according to each model.
- Different colors represent each predictive method for easy comparison.

Purpose: Allows analysts to identify matchups with high prediction uncertainty and compare the performance of individual models.

4.13 Key Differences Across Methods

Feature	Monte Carlo	Elo + Offense	Logistic Regression	Neural N
Game-to-game randomness	High	Moderate	Moderate	Mode
Long-term team strength	Indirect	Direct	Indirect	Indin
Uses scoring distributions	Yes	No	No	No
Uses win/loss history	Indirect	Yes	Yes	Ye
Captures nonlinear relationships	No	No	Limited	Ye

Table 1: Comparison of predictive methods used in the updated simulation framework.

By combining all four methods, the model accounts for both stochastic variability and consistent team performance trends, providing a comprehensive prediction framework.

4.14 Variable Definitions

Simulation Settings:

- `num_simulations` — Number of Monte Carlo trials per matchup (e.g., 50,000).
- `games_to_simulate` — List of team matchups evaluated.

Dataset Variables:

- `homeScore`, `awayScore` — Final game scores.
- `hometeamName`, `awayteamName` — Team identifiers.

Statistical Model Variables:

- `avg_pts` — Average points scored.
- `std_pts` — Standard deviation of points scored.
- `avg_allowed` — Average points allowed.
- `std_allowed` — Standard deviation of points allowed.

Elo Model Variables:

- `elo[team]` — Current Elo rating of a team.
- `K` — Adjustment factor controlling rating changes (set to 20).
- `expected_home` / `expected_away` — Predicted win probabilities before a game.

4.15 Code Implementation Explanation

Data Preparation

The program first loads historical NBA game data from a CSV file. Score columns are converted to numeric values, and incomplete records are removed. Historical team names are mapped to modern franchise names to maintain consistency.

Elo Simulation Procedure

1. Initialize all team ratings at 1500.
2. Process historical games:
 - Compute expected win probability from rating difference.
 - Compare expected outcome to actual result.
 - Update both teams ratings using the formula

$$\text{Rating}_{\text{new}} = \text{Rating}_{\text{old}} + K \cdot (\text{Actual} - \text{Expected})$$

3. Simulate future matchups using random sampling according to calculated win probabilities.
4. Generate bar graphs showing simulated win counts.

Statistical Simulation Procedure

1. Compute team statistics (average points scored, allowed, and standard deviations).
2. Simulate individual games:
 - Generate random offensive scores from normal distributions.
 - Adjust scores using opponent defensive distributions.
 - Determine winner based on higher simulated score.
3. Repeat thousands of times to estimate win probabilities.
4. Generate bar charts for visual comparison.

4.16 Hybrid Interpretation

Agreement between the two methods suggests robust predictions, while disagreement indicates match ups sensitive to assumptions about variability versus overall strength.

5 Multiple Linear Regression

Multiple Linear Regression (MLR) is a statistical method used to model the relationship between a single response variable and multiple predictor variables. By examining how several explanatory variables simultaneously influence an outcome, MLR allows the magnitude and direction of each variable's contribution to be examined. Built on the dataset "NBA Stats (1947 - Present)" found on kaggle, the model is designed to predict a team's winning probability.

5.1 Model Math

MLR Assumptions

1. The true relationship between y and X_i can be represented by a linear relationship
2. $E[u|x] = 0$: Exogeneity. Variables are influenced by factors independent from the model.
3. $Var(x) \neq 0$: No perfect co-linearity. No combination of predictive variables can perfectly predict another predictive variable.

Hypothesis Testing Assumptions

4. $Var(U|X_1, \dots, X_n) = \sigma^2$: Homoskedasticity. The variance in the error term U is independent of the behavior occurring in X_i .
5. $Cov(u_i, u_j) = 0$ for $i \neq j$: No two individuals have correlated error terms

6 Formulas and Calculations

The simulation framework relies on several mathematical formulas and calculations to estimate game outcomes. Each predictive model uses specific formulas to quantify team performance and win probabilities.

6.1 Elo Rating Update Formula

The Elo rating system is used to measure overall team strength. After each historical game, team ratings are updated using:

$$\text{Rating}_{\text{new}} = \text{Rating}_{\text{old}} + K \times (\text{Actual} - \text{Expected})$$

where:

- $\text{Rating}_{\text{old}}$ is the team's rating before the game.
- K is the adjustment factor controlling how much ratings change after a game (set to 20).
- Actual is the game outcome (1 if the team wins, 0 if the team loses).
- Expected is the predicted probability of winning, calculated as:

$$\text{Expected}_{\text{home}} = \frac{1}{1 + 10^{(\text{Elo}_{\text{away}} - \text{Elo}_{\text{home}})/400}}$$

$$\text{Expected}_{\text{away}} = 1 - \text{Expected}_{\text{home}}$$

This formula adjusts team ratings based on performance relative to expectation, capturing long-term team strength.

6.2 Monte Carlo Score Simulation Formula

The Monte Carlo method simulates individual game outcomes using normally distributed random scores:

$$\text{Score}_{\text{team}} \sim \mathcal{N}(\mu, \sigma^2)$$

where:

- μ is the expected points per game (PPG) for the team, adjusted for Elo rating differences.
- σ is the standard deviation of points scored, representing game-to-game variability (approximately 12 points in the model).

The probability of a team winning is estimated as the fraction of simulated games where its score exceeds the opponent's score.

6.3 Elo + Offensive Strength Probability Adjustment

To incorporate offensive efficiency, the win probability from Elo is adjusted based on team scoring relative to league average:

$$P_{\text{win}} = \text{Base_Elo_Probability} \times \frac{\text{Offensive_Rating}_{\text{team}}}{\text{Offensive_Rating}_{\text{opponent}}}$$

where:

- Base_Elo_Probability is the standard Elo win probability.
- Offensive_Rating is the team's points per game divided by the league average PPG.

This adjustment ensures teams with stronger offensive output are more likely to win, even if Elo ratings are similar.

6.4 Logistic Regression Model Formula

The logistic regression model estimates the probability that the home team wins based on differences in team statistics:

$$P(\text{home win}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \Delta \text{win_pct} + \beta_2 \Delta \text{pts_scored} + \beta_3 \Delta \text{pts_allowed} + \beta_4 \Delta \text{PPG})}}$$

where:

- Δ indicates the difference between the home and away team for each statistic.
- β_0 is the intercept term.
- $\beta_1, \dots, \beta_4$ are coefficients learned from historical game data.

This formula captures the relationship between historical team performance metrics and the probability of winning.

6.5 Neural Network Model Formula

The neural network model uses a feed-forward architecture to approximate a complex function mapping team statistics to win probability:

$$\mathbf{h}^{(1)} = f(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)})$$

$$\mathbf{h}^{(l)} = f(\mathbf{W}^{(l)}\mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}), \quad l = 2, 3$$

$$P(\text{team win}) = \sigma(\mathbf{W}^{(L)}\mathbf{h}^{(L-1)} + \mathbf{b}^{(L)})$$

where:

- $\mathbf{x}$ is the input vector of team feature differences.
- $\mathbf{h}^{(l)}$ is the hidden layer output at layer l .
- $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ are the weight matrix and bias vector at layer l .
- f is the ReLU activation function.
- σ is the sigmoid function producing a probability between 0 and 1.
- L is the total number of layers (input, hidden, output).

This formulation allows the model to capture nonlinear interactions between team statistics and predict win probabilities.

6.6 Overall Probability Calculation

For each matchup, the final predicted probability is computed as the average of the four models:

$$P_{\text{overall}} = \frac{1}{4} (P_{\text{Monte Carlo}} + P_{\text{Elo+Offense}} + P_{\text{Logistic}} + P_{\text{Neural Network}})$$

This combined estimate balances the strengths of each predictive method, taking into account both game-to-game variability and long-term team performance trends. (Alloush)

Multiple Linear Regression Equation

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + U_i$$

y: Response variable

$\beta_1 \rightarrow \beta_n$: Slope parameters

β_0 : Referent

U_i : Error Term

1. Response Variable

This variable's true behavior is unknown so the model will make a prediction instead. In our case, the response variable is NBA team win probability.

2. Slope Parameters

Our model contains categorical and numerical variables. This requires the model to process two different types of inputs. While the numerical variables receive simple number inputs, categorical variables use dummy variables. These are simple binary predictors that signal which values of beta that represent NBA teams should be called upon.

Example Dummy Variable

$$X_i = \begin{cases} 1, & \text{Boston Celtics} \\ 0, & \text{otherwise} \end{cases}$$

The interpretation of the slope parameters reveal the nature of the relationship between the predictors and the win probability. For example, BOS has the coefficient $\beta_2 = -2.867e - 02$. This would be interpreted as the Celtics are associated with a 2.87 point decrease in win probability while holding all else constant (Alloush). These parameters can also be paired for joint variable interpretations. For context, β_1 represents basketball season and β_{31} is average roster age. $2026\beta_1 + \beta_2$ is the expected win probability for the 2026 Boston Celtics while holding all else constant. Another example: $2026\beta_1 + \beta_2 + 25.6\beta_{31}$. This would read as the expected win probability of the 2026 Celtics with an average roster age of 25.6 years old while holding all else constant.

3. Referent

Due to the model not having perfect co-linearity, one NBA franchise must be excluded from being assigned a dummy variable input. This excluded value is formally called the referent, and is represented by β_0 (Alloush). This model's referent is ATL or the Atlanta Hawks. It was chosen simply due to the alphabetical ordering of NBA teams in the original data set. β_0 can be interpreted as the expected win probability of the Atlanta Hawks when all other predictors are held at zero. The referent can also be cross compared to other variables in the model. For example, $-\beta_2$ would be the expected difference in win probability between the Atlanta Hawks and Boston Celtics while holding all else constant (Alloush).

4. Error term

The error term represents all influential factors of win probability that are not contained in the model (Alloush). This includes confounding variable influence, which is when a variable causes influence in both the predictor and the response, distorting the true nature of their relationship (Alloush). The error term is excluded and not assigned a value in the practical application of the model. Instead, by not existing, it represents the information and variables left out before the final beta values were calculated.

Beta Derivation

A quick note, ordinary least squares is the process calculating a single β , while multiple linear regression is the combination of them all.

$$Y = X\beta + \varepsilon$$

Y: An $n \times 1$ vector of observed response values.

X: An $n \times (p + 1)$ design matrix containing the predictors, with the first column consisting of ones for the intercept.

β : A $(p + 1) \times 1$ vector of unknown regression coefficients.

ε : An $n \times 1$ vector of error terms.

(Walde, 2025)

1. Define the residuals

$$e = Y - X\beta$$

Residuals represent the difference between observed values and the predicted values produced by the regression model. The goal of ordinary least squares is to make these residuals as small as possible (Walde, 2025).

2. Define the residual sum of squares

$$RSS(\beta) = (Y - X\beta)^T(Y - X\beta)$$

The residual sum of squares (RSS) measures the total squared difference between observed and predicted values. Ordinary Least Squares estimates β by minimizing this quantity (Walde, 2025).

$$RSS = Y^T Y - Y^T X \beta - \beta^T X^T Y + \beta^T X^T X \beta$$

Due to the expansion being a scalar, $Y^T X \beta = \beta^T X^T Y$ (Walde, 2025).

$$RSS = Y^T Y - 2\beta^T X^T Y + \beta^T X^T X \beta$$

3. Differentiate with respect to β

$$\frac{\partial RSS}{\partial \beta} = -2X^T Y + 2X^T X \beta$$

The derivative is taken with respect to the vector of coefficients. This step identifies how RSS changes as the coefficients change (Walde, 2025).

Set the derivative equal to zero

$$-2X^T Y + 2X^T X \beta = 0$$

$$X^T X \beta = X^T Y$$

Setting the derivative equal to zero identifies the minimum of the RSS function. This produces the normal equations (Walde, 2025).

4. Solve for the coefficient vector

$$\beta = (X^T X)^{-1} X^T Y$$

Multiplying both sides by $(X^T X)^{-1}$ isolates the coefficient vector. This formula gives the Ordinary Least Squares estimator used to calculate regression coefficients in multiple linear regression (Walde, 2025).

6.7 Model decision making

1. **Data Assumptions:**

- Teams maintain relatively similar scoring and winning trends after the trade deadline despite rostering new players
- No injuries are suffered by a team's main contributors
- A team's starting lineup is always the same and players are not benched for 'easy' match ups

2. Defining the Modern Era:

The data used to build the model was an in-depth historical database that analyzed the evolution of basketball in the US. The earliest data recorded was from 1947 and continued until the current season. While an extremely large historical database is typically a godsend for a predictive model, to maintain accuracy for the modern league this dataset needed to be dissected. Throughout its history, the NBA has had different play styles and strategies that have dominated the league. As the league has evolved, new successful strategies have emerged beating older approaches. So, it was important to try and define the year that the 'modern league' began. To enable the model predicted applicable results and not a blend of every outdated era of basketball.

Extremely subjective, there is no single moment when each era has come and went. Many basketball enthusiasts have differing opinions on which player redefined a generation of basketball. But, the general consensus is that in the mid-early 2000's a new era of play was born. With this in mind, I decided to use the oldest tenured active player, LeBron James, to be a fair measurement of the modern league. Dominant his entire career, LeBron has been a bastion of the league since he was first drafted in 2004. Always at the center of debate for the best player in the league, I felt his play style's continued success throughout the 2000s and well into the 2020s could be a fair assessment of modern play.

3. Variable selection

Response Variable: Winning Percentage: How often a team wins out of the total number of games played. While team point scoring can also reflect a teams ability to win, the margins are much smaller. Heavy favorites often win by one or two scores, misrepresenting the win probabilities provided by sports books. To undercut the additional model complexity needed to identify the nuanced trends of basketball scoring, general game outcome was chosen instead.

Explanatory Variables:

The following predictors were hand picked based on personal basketball knowledge. With the intent of choosing statistics that had no derivation from the response variable.

- (a) **Abbreviation:** Each team has a designated abbreviation for their respective city that they are from. Ex. The Denver Nuggets are denoted DEN in the data set. This variable was important to include as it represented the importance of team ownership in winning. Some franchises are owned by selfish management, causing their histories to be plagued with losing and bad trades.
- (b) **Season:** Represents the year of play, but due to the season occurring over the course of two years, season corresponds to the year the playoffs for that season occur. Ex, the 2025-2026 season is denoted 2026 as the playoffs occur in May of 2026. Season is important to include as it signifies the scrambling of rosters and hiring of new coaching staff.
- (c) **Age:** An average created using the active roster for the respective team. Age can signify both experience and innovation in finding a way to score. Reflecting a teams overall ability to work together and win.

- (d) **Attend_g**: A measure of how many people on average attend home games for the respective team. Good teams inspire their city with their success. Home game attendance can reflect a teams historical success, hope for the current season, and franchise culture.
- (e) **MOV**: Margin of victory in points for a respective team. Given a random game, what is the average net point score a team would finish with in comparison to their opponent. Ex. -1.1, typically finish a game by losing by 1.1 points to their opponent. Margin of victory is a measure of how dominant a team is. It tests whether a team's wins are merely a fluke or well deserved.
- (f) **Overall pick**: A measure of the highest draft pick or a teams earliest turn to pick a new player that is not currently playing in the NBA. The draft system is designed to balance the league, with the worst teams getting the best chances to get new talent. That said, some teams tank so that they can try and get the best pick possible. Overall pick is a reflection of a teams recent success and current motivation to win.
- (g) **All.Star.Appearances.by.Year**: Given a respective season, a count of the number of players on the same team that are given an in season accolade for exceptional performance.
- (h) **opp_x3p_percent**: Opposition three point shooting percentage.
- (i) **opp_x2p_percent**: Opposition two point shooting percentage.
- (j) **opp_ft_per_game**: Opposition free throws made per game on average. Free throws can represent referee bias. Sometimes the NBA rewards its most marketable teams the most fouls so that they can continue to be successful. Allowing the team to attract new fans to the NBA.
- (k) **x3p_percent**: Three point shooting percentage of target team.
- (l) **x2p_percent**: Two point shooting percentage of target team.
- (m) **ft_per_game**: Free throws per game on average of target team.

4. Data Modifications

In the original dataset, there was one instance of outside data modification. In the NBA, when teams trade for players they can bundle in draft picks. Given teams have a finite number of draft picks per year, it's possible teams shed all of their picks for a single year in trade deals. This causes a simple NA in the dataset, which normally can be settled with a zero. However, when drafting, the lower the number the better a position a team is in. So, I decided to allocate the penalty of 61 to NA values instead. The NBA draft has 60 selections over the course of two rounds. So, a team without any picks in a given year is assumed to have the pick after the absolute last pick, 61 as their first and best chance to draft a new player.

6.8 Influence Testing

Before the model results can be analyzed, it is important to establish that variability in the model predictions are not occurring from pure chance alone. A global F-test can be used to verify the existence of a relationship between the response and predicting variables. The null hypothesis declares that no single variable in the model has any relationship with the response. With the alternative saying otherwise. It is important to note that this hypothesis test does not reveal what the variables of influence are should the null be disproved. It will only give evidence of their existence.

Global F-test

$$H_0 : \beta_0 = \beta_1 = \dots = \beta_n = 0$$

$$H_a : \beta_0 \neq \beta_1 \neq \dots \neq \beta_n \neq 0$$

$$F_{stat} = \frac{\frac{SS_{reg}}{k}}{\frac{SS_{res}}{n-k-1}} = \frac{MS_{reg}}{MS_{res}} = \frac{\frac{13.981}{41}}{\frac{0.76012}{613}} = 275$$

n: Sample Size

k: Total number of predictors, including the intercept β_0

$\hat{y}_i$: Predicted y value

$\bar{y}$: Mean of the response variable

MS_{reg} : Mean Squared Regression

MS_{res} : Mean Squared Residual

(Alloush, 2025)

$$SS_{reg} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

Sum of Squares Regression is the measure of all the variation in the response variable y that can be explained by the model (Alloush, 2025).

$$SS_{res} = \sum (y - \hat{y}_i)^2$$

Sum of Squares Residuals is the measure of the variation in y that cannot be explained by predictive variables (x) (Alloush, 2025).

Critical Values

A simple way to determine the outcome of a hypothesis test is to compare the test statistic with a critical value. The critical value can be found using a simple F-table online. The F statistic table will evaluate three inputs, they are: α level 0.05, df_1 41, and df_2 613. This generates a critical value of 1.4.

For context, α level represents the probability of a type one error (Keller). The rejection of a H_0 hypothesis when it is true. Opposite to α is β , representing the probability of a type two error (Keller). The failure to reject a false H_0 hypothesis. Should alpha be too small, the probability of a type two error increases and vice versa. This is where Power comes in. Power is the measurement of α and β 's relationship. Found with the formula $1-\beta$, power represents the probability of correctly rejecting a false hypothesis (Alloush, 2025). Luckily, optimal power has already been determined, with the best levels to run a hypothesis test at being $\alpha = 0.05$ and $\beta = 0.8$ (Alloush, 2025).

df_1 and df_2 represent degrees of freedom, which determine the shape of the F-distribution used in hypothesis testing (Keller). In multiple regression, df_1 corresponds to k , the denominator associated with SS_{reg} . df_2 corresponds with $n - k - 1$, the denominator associated with SS_{res} . These degrees of freedom are used to compute the mean squares MS_{reg} and MS_{res} , whose ratio forms the F-statistic.

Hypothesis Test Decision Making

$$F_{stat} \leq F_{critical} : \text{Fail to reject } H_0$$

$$F_{stat} > F_{critical} : \text{Reject } H_0$$

Global F test conclusion: Given the critical value $1.4 < 275$, we can safely reject the null hypothesis at the 0.05 alpha level. This means that there is sufficient statistical evidence supporting the notion the model is responsible for explaining variation in the response variable.

T-Test

Now that the model has been proven to be operating on more than chance alone, the variables of influence must be identified. The null hypothesis below states β_i has no influential relationship with the response variable.

T-test Hypotheses

$$H_0 : \beta_i = 0$$

$$H_a : \beta_i \neq 0$$

Test Statistic

The test statistic used to prove or disprove H_0 will be a t-stat representing a t-distribution.

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$\bar{x}$: Sample Mean

μ : True mean, but when testing a hypothesis the value H_0 is set equal to is used

n : Sample Size

s : Sample Standard deviation

T-test Results

Once a t-stat is calculated, a two tailed critical value can be found using a t-table. Applying this test to all our variables, 10 are found to be significant. This means they can be used to explain variation in the model instead of chance alone. The variables below are ranked by their t-values, with the largest t-values representing having the most supporting evidence against H_0 . The list below is ordered in descending value:

- MOV
- All Star Appearance by Year
- Overall Pick
- MIN
- Age
- BOS
- NYK
- DET
- Attend_g
- TOR

7 Kelly Criterion O/U

The central problem is:

How should a bettor allocate capital across bets in order to maximize long-term growth while managing risk?

7.1 Comparison of Betting Strategies

We consider the problem of repeated betting on NBA over/under point totals. For illustration, consider hypothetical odds of -100 on both sides of a total (sportsbooks do not typically offer this pricing). Under -100 odds, a bettor risks \$500 to win \$500. The implied probability of success under these odds is

$$p_{\text{sportsbook}} = \frac{500}{500 + 500} \approx 0.5.$$

This method is used to translate sportsbook odds into implied probabilities.

In this portion of the project we analyze two capital allocation strategies under repeated NBA over/under wagers:

1. **Flat betting:** Wager a fixed amount of \$500 on each qualifying bet regardless of edge.
2. **Kelly betting:** Wager the Kelly-optimal fraction

$$f^* = \frac{bp - (1 - p)}{b},$$

(Oberhuber, 2024)

where b is the net payout multiple from sportsbook odds, p is the estimated probability of success derived from Kalshi prices, and $q = 1 - p$.

The central question is:

When two markets provide different probability estimates, how should capital be allocated to balance long-term growth and risk?

This problem is interesting because positive expected value does not guarantee consistent growth. The randomness of outcomes introduces volatility, and poor bet sizing may lead to large drawdowns or ruin. The tradeoff between growth and risk makes this an applied stochastic optimization problem.

7.2 Mathematical Motivation for the Kelly Criterion

The Kelly Criterion arises from maximizing the expected logarithmic growth rate of wealth over repeated bets. Let B denote the current bankroll and suppose a bettor wagers a fraction f of their bankroll on a bet with probability p of winning and net odds b .

If the bettor wins, the bankroll becomes

$$B_{\text{win}} = B(1 + fb),$$

and if the bettor loses, the bankroll becomes

$$B_{\text{loss}} = B(1 - f).$$

The expected logarithmic growth of the bankroll is therefore

$$G(f) = p \log(1 + fb) + (1 - p) \log(1 - f).$$

The Kelly strategy chooses f to maximize this expected growth. Differentiating and solving

$$\frac{dG}{df} = 0$$

yields

$$f^* = \frac{bp - (1 - p)}{b}.$$

(Oberhuber 2024)

This fraction maximizes the long-run exponential growth rate of wealth under repeated independent bets. If $f^* \leq 0$, the bettor should not place the wager, since the expected growth rate would be negative.

7.3 The Method

In this portion of the project, we simulate repeated betting on NBA over/under markets under the two strategies described above.

For each bet:

1. Initialize bankroll at

$$B_0 = \$10,000.$$

2. For each qualifying bet over a 10-day period:

- Use the estimated probability p derived from Kalshi prices.
- Apply sportsbook net odds corresponding to the posted moneyline.
- Update the bankroll according to the selected betting strategy.

3. The expected value per wager can be computed as

$$EV = W(bp - (1 - p)),$$

where W is the wager amount.

4. Record the final bankroll after each day.

Over the course of 10 days, two hypothetical wagers are placed on each game played that day. One wager uses the flat betting strategy of \$500 per game, while the other uses the Kelly Criterion when an edge exists.

The edge is determined by comparing the sportsbook implied probability with the estimated probability derived from Kalshi prices. If the Kalshi probability exceeds the sportsbook implied probability, the bettor is considered to have a positive edge and a Kelly wager is placed.

After each day, the bankroll is updated based on wins and losses. Under the Kelly strategy, the fraction wagered is recalculated using the updated bankroll rather than the original \$10,000.

By comparing strategies, we evaluate:

- Long-term growth potential
- Volatility of outcomes
- Risk-adjusted performance

This approach allows us to analyze how capital allocation decisions interact with probability estimation in realistic betting markets.

7.4 Variables

- $B_0 = 10,000$: Initial bankroll
- p : Estimated true probability (from Kalshi)
- b : Net sportsbook payout multiple
- f^* : Kelly fraction

$$f^* = \frac{bp - (1 - p)}{b}$$

- $W = f^*B$: Optimal wager size
- B_{final} : Final bankroll after event resolution

8 Findings

Multiple Linear Regression

After the implementation of both the Kelley criterion and flat betting, the results are striking. Simple by design, flat betting created far more opportunities for the linear model to find success. Applying a simple 500 flat bet to every NBA game, the model was not burdened with finding an edge. Allowing for a great many more opportunities to bet every night. Over the course of the 10 day exposure period lasting from February 19th to the 28th, 76 games were played. In this period, the flat betting wagered \$38,000 with an average of \$3800 dollars per night. In contrast, the Kelley criterion detected an edge just 19 times. With only about 25% of the same opportunities, Kelley margins were much smaller. Of the games bet on, 10 or 52.63% of the bets hit. A total of \$26,421 was wagered with an average \$2,642.10 per night.

While the betting opportunities differed due to strategy, the model remained constant. Across the 76 games, the model successfully predicted the outcome of 55 games, producing 72.37% accuracy. Due to the Kelley criterion often refusing to produce a bet despite a correct win prediction, the flat betting strategy is more reflective of the models accuracy. Flat betting saw a net profit margin of 34.27%, having accumulated a total net winnings of \$13,022.49. These winnings were calculated using the odds produced on OddsPortal.com, a website that evaluates the odds of various sports books to generate odds reflective of average offerings. The Kelly criterion was not as successful, generating a net earnings of -\$432.21. Thus, yielding a profit margin of -1.64%. The drastic difference in earnings can be linked to the sizable difference in bet sizes. Flat betting is safe, easily diversifying risk across many games. While Kelly is much more aggressive, placing bets as large as \$6865 on a single game. So, even though more bets hit than missed, most money was won by betting on favorites. This caused the bets to be much smaller as the edge was small. The losses however were large, as any opportunity for an upset required a sizable bet.

Simulations

For the model using simulations, the NBA game outcome predictor was evaluated using historical game results, team season totals, and team statistics from 1947 to the present. The

predictive framework combined four approaches: Monte Carlo simulations generating 50,000 game score scenarios per matchup with Elo-adjusted distributions, Elo ratings adjusted for offensive efficiency, logistic regression trained on recent team statistics, and a neural network modeling team win probabilities.

The Monte Carlo component simulates individual game scores by drawing from normal distributions based on each team’s historical points per game and scoring variability, while incorporating Elo-based adjustments to account for relative team strength. The Elo component quantifies overall team quality, updating ratings sequentially after each historical matchup using a K-factor of 20. These ratings are then used to calculate expected win probabilities, reflecting long-term team consistency independent of individual game variability.

To extend predictive power, logistic regression was trained using historical team performance metrics, including win percentage, points scored, points allowed, and points per game, capturing linear relationships between team performance and game outcomes. A neural network with a multi-layer perceptron architecture was also trained on the same features, allowing for non-linear patterns and interactions to inform predictions. Team statistics were standardized and missing data handled with league averages to ensure model stability across all matchups.

The Python implementation begins by loading historical game data, team totals, and seasonal statistics. Legacy team names were mapped to current franchises for consistency, and missing or invalid data were cleaned. Elo ratings were initialized at 1500 for all teams and updated game-by-game according to actual results compared to expected outcomes, producing dynamic team strength ratings. Monte Carlo simulations were performed with 50,000 trials per matchup, generating probabilities that each team would win based on simulated scores. Logistic regression and neural network models were trained on historical data, with features representing the differences in team performance metrics between opponents, and then used to generate predicted win probabilities for each matchup.

For prediction, the model calculated four separate probabilities per game: Monte Carlo score simulations, Elo-adjusted win probability, logistic regression prediction, and neural network prediction. The final prediction for each matchup is the average of these four methods, balancing short-term randomness with long-term performance trends. Expanding the analysis to the full month of February 2026, the multi-model NBA predictor was evaluated across 76 games from February 1st to the 28th. Using a consistent \$312.50 flat wager per game, the model correctly predicted 50 of 76 games, corresponding to an overall accuracy of 65.8%. This approach would have generated \$15,625 in gross winnings, \$8,125 in losses, and a net profit of \$7,500 for the month, demonstrating both the predictive accuracy of the hybrid model and the practical effectiveness of a systematic flat betting strategy.

Kelly Criterion

Through ten days of observation and imaginative betting, the results came to a surprise. However, as hoped, the kelly criterion saw a greater success rate than flat betting. Out of the total 75 games only 26 looked to have an edge on the O/U betting and were then bet on using the Kelley method. The final bankroll for the flat betting was \$7,284.49 losing approximately 27.16% of the initial value, while the Kelley method’s final was \$12,404.20 gaining a approximately 24.04%. The results of our ten-day NBA betting experiment are summarized in Table 2. The table reports cumulative bankroll changes for both flat betting and Kelly Criterion strategies.

Strategy	Starting Bankroll	Ending Bankroll
Flat Betting	\$10,000	\$7,284.49
Kelly Criterion	\$10,000	\$12,404.20

Table 2: Cumulative bankroll results for flat betting versus Kelly Criterion betting over ten days of NBA games.

Day-to-Day Observations

- The flat betting strategy experienced several losing streaks, resulting in a net loss of \$2,715.51.
- Kelly Criterion betting produced consistent gains by adjusting wager sizes according to edge, resulting in a net gain of \$2,404.20.
- Certain games with high perceived probability (e.g., odds of -850 or -521) contributed disproportionately to Kelly gains due to larger wagers relative to bankroll.
- Flat betting does not exploit positive edge, so even games with a strong likelihood of success only produced standard profits or losses of \$500.

Example Kelly Computation

Consider a game with moneyline odds -112 and estimated winning probability $p = 0.53$:

$$b = \frac{100}{112} \approx 0.8929, \quad q = 1 - 0.53 = 0.47$$

$$f^* = \frac{0.8929 \cdot 0.53 - 0.47}{0.8929} \approx 0.064 \text{ (6.4\% of bankroll)}$$

$$\text{Expected Profit} = 0.064 \cdot 10000 \cdot (0.53 \cdot 0.8929 - 0.47) \approx 446.43$$

This matches the observed profit for that game in our dataset.

Sensitivity Analysis

- Small errors in estimating the probability p ($\pm 1 - 2\%$) have limited impact on the expected gains for low-edge bets.
- Large errors in probability estimation can lead to overbetting (if p is overestimated) or missed opportunities (if p is underestimated).
- The Kelly strategy dynamically adjusts wagers as bankroll changes, so the strategy naturally scales risk according to cumulative performance.

Discussion

- The Kelly Criterion not only increases expected returns but also reduces volatility compared to flat betting by proportionally sizing bets.
- Even with modest edges, Kelly betting can grow the bankroll over time, whereas flat betting can quickly deplete capital during negative streaks.
- Practical limitations include: estimation errors for p , the presence of sportsbook vig, and potential maximum bet limits which can constrain the theoretical Kelly optimal bet.
- These results illustrate the mathematical advantage of edge-based proportional betting in real-world sports wagering.

Summary Statistics

To further quantify the performance of each strategy, we computed daily profit/loss statistics, including the mean, median, standard deviation, and probability of substantial drawdowns. For this analysis, we define a *substantial drawdown* as a daily loss exceeding \$500 (5% of the starting bankroll).

Statistic	Flat Betting	Kelly Criterion
Mean Daily Profit/Loss	-\$68.13	\$292.74
Median Daily Profit/Loss	-\$175.34	\$246.06
Standard Deviation	\$967.12	\$879.16
Probability of Drawdown (\$500 loss)	44.4%	10%

Table 3: Summary statistics of daily profit/loss for Flat Betting and Kelly Criterion strategies over ten days of NBA games.

Discussion of Summary Statistics

The summary statistics provide several insights:

- **Flat betting** exhibits slightly negative mean daily profit and high volatility, with nearly half of the days resulting in substantial drawdowns. This illustrates the risk of using fixed-size bets without regard to the edge of each wager.
- **Kelly Criterion** shows a positive mean and median daily profit, lower probability of large losses, and slightly lower volatility compared to flat betting. This highlights the effectiveness of proportional betting based on estimated edge.
- The lower probability of substantial drawdowns under the Kelly strategy demonstrates its risk-management advantage: by adjusting bet sizes dynamically according to bankroll and edge, it mitigates the impact of losing streaks while capitalizing on favorable opportunities.

- Overall, the statistics quantitatively confirm that Kelly Criterion betting outperforms flat betting both in expected growth and in controlling risk.

9 Project Questions

A major limiting factor for this project was accessing relevant and accurate data. Due to the gambling market having grown quite large since its legalization, many data sources hide behind pay walls. But is this data really needed to build an accurate model? It would be interesting to understand what additional value this data could add. One valuable feature could be historical odds. This could allow us to evaluate the overall accuracy of book makers and adjust accordingly.

10 Future Areas of Study

In the next few weeks the basketball post season will begin and with that March Madness. March Madness has been a well documented national attraction for years. This would provide lots of opportunities to find in-depth data documenting each tournament. Additionally, a large part of the attraction of the tournament is filling out a bracket before hand and trying to guess the winner. Every year people post online about their strategies and decision making in picking the right bracket. But no matter how self-assured someone may appear, they are rarely correct. Some of the people that find major success do not watch basketball at all. With their game predictions being motivated by arbitrary factors like mascots or team colors. There is a large realm of things that could be studied but the most pressing is team seeding. Seeding is a rank from 1-15 assigned to a team to determine how favored they are to win. The tournament is divided by region, so four teams hold each seed. But more importantly, upsets happen, coining the term a Cinderella run. Which is when a low seeded team (underdog) goes on a winning streak beating out teams declared to be much better than them. Seeding is speculative and is based on regular season conference difficulty and game record. Thus begging the question, could we seed the teams ourselves and make an accurate bracket prediction?

11 Appendix

Note: The city abbreviations represent dummy variables corresponding each to NBA team.

$$\begin{aligned}\beta_0 &: 4.718e - 02 \\ \beta_1 &: 1.990e - 04 \text{ Season} \\ \beta_2 &: -2.867e - 02 \text{ BOS} \\ \beta_3 &: -1.325e - 03 \text{ BRK} \\ \beta_4 &: -1.430e - 03 \text{ CHI} \\ \beta_5 &: -1.953e - 03 \text{ CHO} \\ \beta_6 &: 2.016e - 03 \text{ CLE} \\ \beta_7 &: -4.569e - 03 \text{ DAL} \\ \beta_8 &: 1.048e - 02 \text{ DEN}\end{aligned}$$

$\beta_9 : -2.313e - 02$ DET
 $\beta_{10} : -1.648e - 02$ GSW
 $\beta_{11} : -8.624e - 03$ HOU
 $\beta_{12} : -5.038e - 03$ IND
 $\beta_{13} : -1.306e - 02$ LAC
 $\beta_{14} : -1.276e - 03$ LAL
 $\beta_{15} : 5.441e - 03$ MEM
 $\beta_{16} : -1.288e - 02$ MIA
 $\beta_{17} : -1.142e - 02$ MIL
 $\beta_{18} : -3.771e - 02$ MIN
 $\beta_{19} : -1.724e - 02$ NOP
 $\beta_{20} : -2.877e - 02$ NYK
 $\beta_{21} : -4.125e - 03$ OKC
 $\beta_{22} : -1.036e - 02$ ORL
 $\beta_{23} : -8.122e - 03$ PHI
 $\beta_{24} : -2.037e - 03$ PHO
 $\beta_{25} : 1.268e - 02$ POR
 $\beta_{26} : -9.143e - 03$ SAC
 $\beta_{27} : -5.284e - 03$ SAS
 $\beta_{28} : -2.223e - 02$ TOR
 $\beta_{29} : -7.607e - 0$ UTA
 $\beta_{30} : -9.049e - 03$ WAS
 $\beta_{31} : 3.781e - 03$ age
 $\beta_{32} : 8.893e - 07$ attend_g
 $\beta_{33} : 2.785e - 02$ mov
 $\beta_{34} : 3.622e - 04$ overall_pick
 $\beta_{35} : 1.175e - 02$ All.Star.Appearance.by.Year
 $\beta_{36} : -2.131e - 01$ opp_x3p_percent
 $\beta_{37} : 1.965e - 02$ opp_x2p_percent
 $\beta_{38} : -3.739e - 04$ opp_ft_per_game
 $\beta_{39} : 2.338e - 02$ x3p_percent
 $\beta_{40} : -3.359e - 02$ x2p_percent
 $\beta_{41} : 4.804e - 04$ ft_per_game

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